DO NOW — RECALL

Network security, proxies and firewalls

Name: _____ Class: _____ Date: _____

1. Can network packets travelling between the same two endpoints take different routes?

2. What is the transport-layer role of TCP?

3. What does a rule ‘allow if staff AND authenticated’ mean if authenticated is false?

4. Why can an open port be useful but also increase risk?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Yes, routing decisions and changing network conditions can lead to different paths.

2. Model answer

A reliable, ordered byte stream between endpoints.

3. Model answer

Deny; AND requires both conditions true.

4. Model answer

It permits needed service connections, but exposes that service to network-originated attacks or probing.

Vocabulary — know these terms

firewall — Security control that permits or blocks network traffic according to rules.

forward proxy — Intermediary that makes requests on a client's behalf.

filter rule — A condition and action defining whether traffic is permitted.

cache — Stored copy of suitable data for reuse.

defence in depth — Use multiple complementary security controls rather than trusting one.

NEW LEARNING

Read, annotate and explain

Check a firewall rule

A firewall applies configured rules to traffic according to properties such as direction, source/destination, ports and state. A basic rule example can allow TCP port 443 but deny 22.

Different request, different decision

A TCP connection to port 22 is denied by the illustrated policy. A firewall rule is not automatically a guarantee that the permitted service is safe.

A proxy is an intermediary

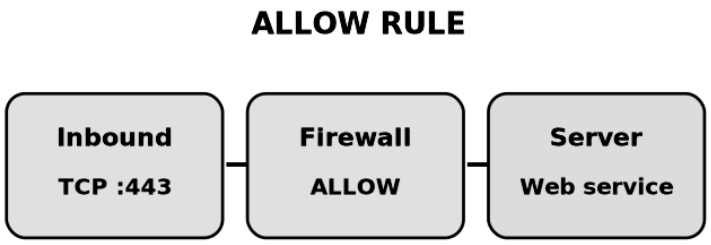
A forward proxy makes requests on behalf of clients, potentially enforcing policy or caching suitable resources. HTTPS content remains encrypted unless a controlled interception mechanism is in place.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

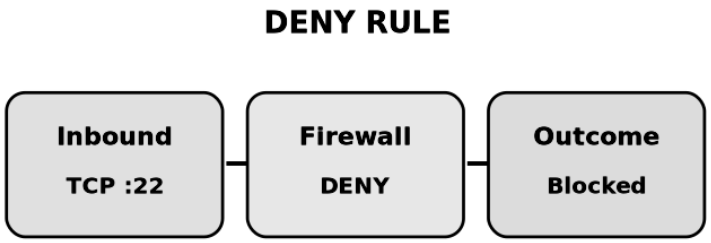
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



ALLOW RULE. Inbound: TCP :443; Firewall: ALLOW; Server: Web service

New Learning

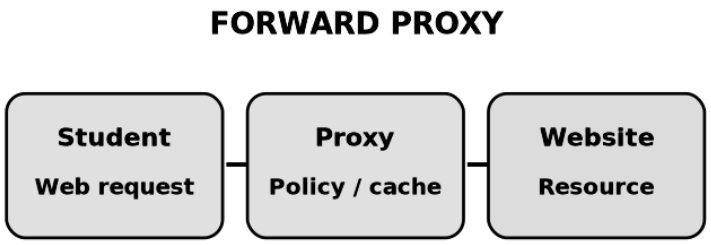


DENY RULE. Inbound: TCP :22; Firewall: DENY; Outcome: Blocked

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



FORWARD PROXY. Student: Web request; Proxy: Policy / cache; Website: Resource

I DO — WORKED EXAMPLE

Follow the worked method

For an inbound rule set allowing TCP 443 and denying TCP 22, decide whether two connection requests are permitted.

443

An inbound TCP connection destined for port 443 matches the allow rule and is permitted.

22

An inbound TCP connection destined for port 22 matches the deny rule and is blocked.

Scope

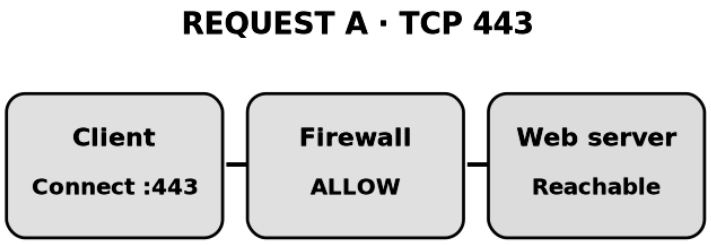
An allow rule permits traffic; it does not prove that an application or connection is harmless.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

For an inbound rule set allowing TCP 443 and denying TCP 22, decide whether two connection requests are permit

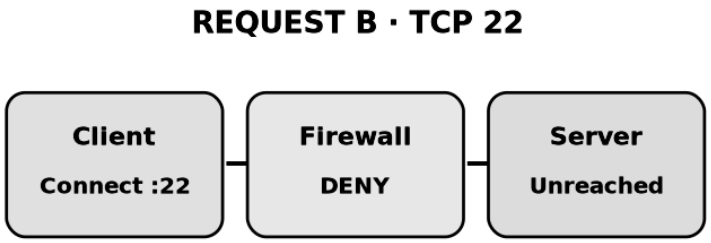
An inbound TCP connection destined for port 443 matches the allow rule and is permitted.



REQUEST A · TCP 443. Client: Connect :443; Firewall: ALLOW; Web server: Reachable

For an inbound rule set allowing TCP 443 and denying TCP 22, decide whether two connection requests are permit

An inbound TCP connection destined for port 443 matches the allow rule and is permitted.



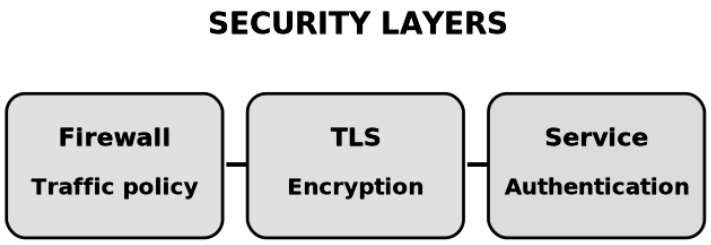
REQUEST B · TCP 22. Client: Connect :22; Firewall: DENY; Server: Unreached

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

For an inbound rule set allowing TCP 443 and denying TCP 22, decide whether two connection requests are permit

An inbound TCP connection destined for port 443 matches the allow rule and is permitted.



SECURITY LAYERS. Firewall: Traffic policy; TLS: Encryption; Service: Authentication

WE DO — GUIDED PRACTICE

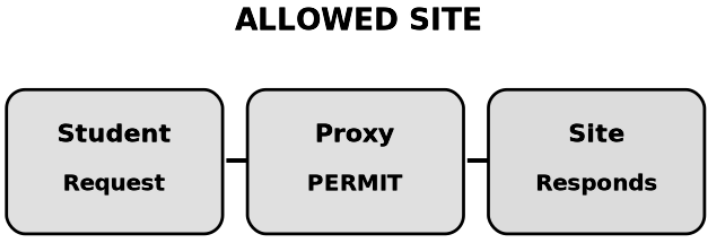
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school uses a forward proxy to restrict browsing. Explain the route for a permitted request and for a blocke

The student's request reaches the proxy, which permits it and contacts the website; a suitable cached response may be used instead.



ALLOWED SITE. Student: Request; Proxy: PERMIT; Site: Responds

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Allowed resource The student's request reaches the proxy, which permits it and contacts the website; a suitable cached response may be used instead.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

A firewall allows inbound TCP 443 but denies inbound TCP 22. Is an inbound connection to destination port 22 allowed? Explain. [2]

Question 2

Name and explain two different functions of a forward proxy in a school network. [4]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Why is the claim ‘a firewall makes all allowed traffic safe’ incorrect? Explain two risks that can remain on allowed connections and identify two additional safeguards. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A forward proxy and firewall are identical because both can restrict access.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

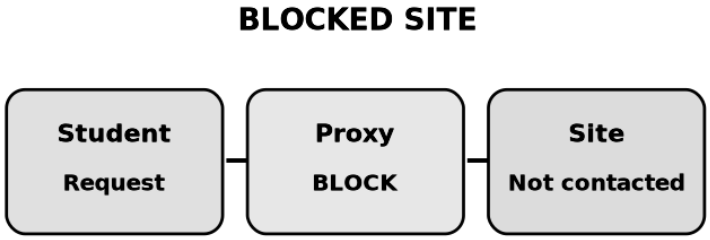
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school uses a forward proxy to restrict browsing. Explain the route for a permitted request and for a blocke

The student's request reaches the proxy, which permits it and contacts the website; a suitable cached response may be used instead.



BLOCKED SITE. Student: Request; Proxy: BLOCK; Site: Not contacted

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Blocked resource

If the proxy's policy denies the destination, it does not forward the request to that site; the student receives a blocking response.

IMPROVE: Explain one correction to your own reasoning.